

ERRATA OF THE DISSERTATION

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LAST MODIFIED ON OCTOBER 1, 2026

I found errors in my dissertation after publication. It would be best to catch them before the ication. Alas, I was incautious about catching errors. Hence, I attach the list of errors in my dissertation. This list is not exhaustive, and there may be other errors.

Feel free to contact me if you find any suspicious parts in my dissertation.

- (1) p158–, Theorem 3.3.19: Its proof is incorrect. The main reason is that there is no guarantee that the dagger (that marks a formula) may not move to the precedent, and reside in the succedent forever. The new proof will be available in the upcoming version of [1].
- (2) p183, Definition of $\varrho_\alpha(\sigma)$:
 - $\bar{I}_\Phi(S^n 0)$ is an antecedent... $\rightarrow \bar{I}_\Phi(S^n 0)$ *appears in* an antecedent...
 - $\bar{I}_\Phi$ is a succedent... $\rightarrow \bar{I}_\Phi(S^n 0)$ *appears in* a succedent...
 (A new version of [1] presents a slightly different definition.)
- (3) p188. Proof of Proposition 3.5.1, The definition of $T(X)$.
 - (4): “For every $i, j < |f|$ ” $\rightarrow$ “For every $i < j < |f|$ ”
 - (6): “For $i, j < |g|$ ” $\rightarrow$ “For $i < j < |g|$ ”
- (4) p306, Proposition 5.1.8: The raw union $\bigcup_{i \in \mathbb{N}} \pi^P(x_i)$ is not precisely the thread predendrogram. (See the modified Chapter E.) We take the completion to $\bigcup_{i \in \mathbb{N}} \pi^P(x_i)$ to have a thread predendrogram. The following calculation also requires taking a completion operator accordingly. Also, in the last line of the proof, $\int^2$ is a typo of $\int^1$. Thus, for example, $\int^2 D$ must be $\int^1 D$.
- (5) p459, Example D.1.2: d has length 1, and not 0.
- (6) p472, Theorem D.2.4 and p476, Proposition D.3.1: Replace the expression ‘predendrogram over X ’ by just a ‘predendrogram.’ This is a remnant of an older notion of a predendrogram where every node comes from a fixed linear order X .
- (7) p482, Theorem D.3.11: The current statement is incorrect and requires modification. We need to add the following assumption to the statement:

Suppose moreover that whenever $y \in D$, $x_0 \in E_0$, and $x_1 \in E_1$ satisfy

$$E_0 \models f_0(y) \multimap x_0 \quad \text{and} \quad E_1 \models f_1(y) \multimap x_1,$$

we have $\pi^{E_0}(x_0) = \pi^{E_1}(x_1)$.

This condition ensures that, after identifying D with its images in E_0 and E_1 , all immediate successors of a common node have the same priority order in the amalgamation. With this additional assumption, the amalgamation construction in the proof works as intended.

The statement and proof also contain several minor errors and omissions.

- (a) In the final sentence of the statement, “ $f_0(x)$ is a terminal node in E_1 ” should be “ $f_0(x)$ is a terminal node in E_0 .”
- (b) After constructing the local comparison relation on $E_0 \cup E_1$, the proof should verify that the resulting structure satisfies the terminal-descendant condition for predendrograms. Let $z \in E_0$ and choose a terminal descendant t of z in E_0 . If t remains terminal in the amalgam, there is nothing to prove. Otherwise, $t \in D$ and t is non-terminal in E_1 . Choose a terminal descendant s of t in E_1 . Since t is terminal in E_0 , no proper descendant of t in E_1 can belong to $D = E_0 \cap E_1$. Hence $s \notin D$ and therefore remains terminal in the amalgamation. The argument for nodes of E_1 is symmetric.
- (c) In the last paragraph of the proof, the same argument showing that E_0 is a subpredendrogram of the amalgamation should also be applied symmetrically to E_1 .

- (8) Proposition D.3.13: The ω -homogeneity statement is incorrect as stated. The extension property has to be restricted to compatible extensions. After the three bullets in item (1), add the following assumption:

Suppose moreover that whenever $x \in D$ has an immediate successor y in E , there is an immediate successor z of $f(x)$ in $\mathbb{Q}_1$ such that $\pi^E(y) = \pi^{\mathbb{Q}_1}(z)$.

Equivalently, whenever a node of D is extended in E , its image under f must already be non-terminal in $\mathbb{Q}_1$, and the immediate successors on the two sides must have the same priority order. Under this additional assumption, the asserted quasimorphism $f': E \rightarrow \mathbb{Q}_1$ exists.

Also, there is a minor typo in the statement of the proposition: “ $f': C \rightarrow \mathbb{Q}_1$ ” should be “ $f': E \rightarrow \mathbb{Q}_1$.”

The construction of $\mathbb{Q}_1$ in the proof also requires modification. In clause 1 defining its field, replace the condition “ $q_i \in A_{a_i,0}$ for each $i < m$ ” by the following:

- $a_0 = \emptyset$.
- For each $i < m$, $q_i \in A_{a_{i+1},0} \cup A_{a_{i+1},1}$ while retaining $q_m \in A_{a_m,0} \cup A_{a_m,1}$.

Thus the corrected clause reads:

The field of $\mathbb{Q}_1$ is the set of $\langle q_0, \dots, q_m \rangle$ such that there are $a_0, \dots, a_m \in \mathbb{P}_0$ with

$$a_0 = \emptyset \quad \text{and} \quad a_0 \multimap a_1 \multimap \dots \multimap a_m,$$

such that $q_i \in A_{a_{i+1},0} \cup A_{a_{i+1},1}$ for every $i < m$ and $q_m \in A_{a_m,0} \cup A_{a_m,1}$.

The fourth argument should be modified accordingly. Enumerate

$$E \setminus D = \{u_i \mid i < N\}$$

so that the immediate predecessor of any u_i , if it belongs to $E \setminus D$, occurs earlier in the enumeration. When choosing the image of u_i , let $a = \pi^E(u_i)$. If u_i is terminal, choose its last rational coordinate from $A_{a,1}$. If u_i is non-terminal and its immediate successors have priority order b , choose its last rational coordinate from $A_{b,0} \cup A_{b,1}$. These sets are dense, so the coordinate can be chosen in the required local interval while avoiding the prescribed finite set.

If $x \multimap^E u_i$, the proof must also allow x to be one of the nodes u_j added at an earlier stage, rather than only a node of D . If $x \in D$, the compatibility assumption above ensures that the immediate successors of $f(x)$ have priority order $\pi^E(u_i)$. If $x = u_j$ for some $j < i$, this follows from the choice made when u_j was embedded.

With these modifications, the recursive construction yields the required predendrogram quasimorphism $f': E \rightarrow \mathbb{Q}_1$ with the terminal preservation condition.

The universality assertion remains valid, but it no longer follows immediately from unrestricted ω -homogeneity. To see it, write a countable predendrogram C as $C = \bigcup_{n < \omega} C_n$, where each C_n is a finite subpredendrogram and $C_n \subseteq C_{n+1}$. Such a sequence may be obtained by choosing, for the first finitely many nodes of an enumeration of C , terminal descendants and taking all of their predecessors. Starting with an embedding of C_0 into $\mathbb{Q}_1$, repeatedly apply the corrected extension property. Since the inclusions $C_n \subseteq C_{n+1}$ and the embeddings already constructed preserve terminal nodes, the compatibility condition is automatic. The union of these embeddings is an embedding of C into $\mathbb{Q}_1$.

- (9) Chapter E contains a wide range of errors (in particular, E.6–E.7). I attach a new version of Chapter E separately. Here is the main issues in Chapter E:

- (a) The proof of Theorem E.5.9 was incorrect (the result is still correct.)
- (b) Section E.6 was incorrect. The main issue happens when we define $\varepsilon_{s,t}^P$. The previous definition implicitly assumes that there is a $\mathbf{p}^P(s,t)$ -congruent arity diagram, but it may not exist. Accordingly, we allow $\varepsilon_{s,t}^P = 0$ when there is no $\mathbf{p}^P(s,t)$ -congruent arity diagram.
- (c) Continuing from the previous item, the local comparison relation $\leq$ may no longer be total even over the set of children of a given node. When $\varepsilon_{s,t}^P = 0$, the corresponding $\mathbf{p}^P(s,t)$ -cells for s and t are incomparable. This does not affect the functorialization of a 2-predendrogram since the incomparability happens only when there is no $\mathbf{p}^P(s,t)$ -congruent arity diagram.
- (d) The local well-foundedness of a 2-predendrogram also requires a modification since $\leq$ accordingly. The thread condition on a 2-predendrogram is also changed.

REFERENCES

- [1] Hanul Jeon. *Proof-theoretic dilator and intermediate pointclasses*. 2025. arXiv: [2501.11220](#).
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